

Learning Experience F.1

Ethan Davidson and Patrick Rooney

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Design Changes

Overall, we did not make any major design changes to our function unit from Learning Experience C.3. One design change we made was increasing the number of wires from eight to sixteen for our wire vectors. We also changed how our shift functions work for rem8 and mult4 functions by changing which bits were moved to allow all sixteen bits to be affected by the shift. We also added eight full adders and sixteen muxes, so our adder circuit would be able to handle 16-bit values. The reason we made these design changes was to allow our function unit to handle two 16-bit operands and output a 16-bit result based on a given operation performed by the function unit.

Evaluation of Implementation

Transistor Count

Function Unit Circuit									
Gates / Elements	2AND	3AND	2OR	4x1 Mux	2x1 Mux	NOT	3OR	4AND	16AND
Number of Gates	1	5	2	1	3	12	1	2	1
Transistor Count	6	40	12	62	60	24	8	20	34
Total Transistors	266								

Figure 1.1 Function Unit Transistor Count

Arithmetic Block Full Circuit			
Gates / Elements	XOR	4x1 Mux	Full Adder
Number of Gates	1	26	16
Transistor Count	12	1612	1152
Total Transistors	2776		

Figure 1.2 Arithmetic Block Transistor Count

Logic Block Full Circuit					
Gates/ Elements	8x1 mux	NOT	NOR	AND	NAND
Number of Gates	1	2	1	1	1
Transistor Count	104	4	4	6	4
Total Transistors	122				

Figure 1.3 Logic Block Transistor Count

Shift Block Full Circuit	
Gates/Elements	2x1 Mux
Number of Gates	1
Transistor Count	20
Total Transistors	20

Figure 1.4 Shift Block Transistor Count

Busy System Transistor Count				
Circuit Element	3AND	2OR	NOT	4x1 mux
Number Of Elements	4	4	2	2
Transistor Count	32	24	8	124
Total Transistors	188			

Figure 1.5 Bus System Transistor Count

Total Circuit Transistor Count					
Circuit Element	Function Unit	Arithmetic Block	Logic Block	Shift Block	Bus System
Number Of Elements	1	1	1	1	1
Transistor Count	266	2776	122	20	188
Total Transistors	3372				

Figure 1.6 Total Circuit Transistor Count

Propagation Delay

Total Circuit Propogation Delay						
Gates/ Elements	NOT	4x1 Mux	Full Adder To Carry	Full Adder to Sum	2OR	3AND
Number of Gates	2	3	15	1	1	1
Propogation Delay (ps)	50	450	2625	250	75	100
Total Propogation Delay (ps)	3550					

Figure 1.7 Total Circuit Propagation Delay

Our final circuit consisted of a circuit for the function unit, an arithmetic block, a logic block, a shift block, and a bus system. Our transistor counts for the function unit circuit (*Figure 1.1*), arithmetic block (*Figure 1.2*), logic block (*Figure 1.3*), the shift block (*Figure 1.4*), and the bus system (*Figure 1.5*) were all added together to find the total transistor count shown in *Figure 1.6*. It is important to note that we had previously calculated that a 2x1 mux has 20 transistors, a 4x1 mux has 62 transistors, and a 8x1 mux has 104 transistors. These ‘assumptions’ are shown in our calculations of final transistor counts. We also assumed that every 2 transistors caused a 25 ps propagation delay.

Simulation

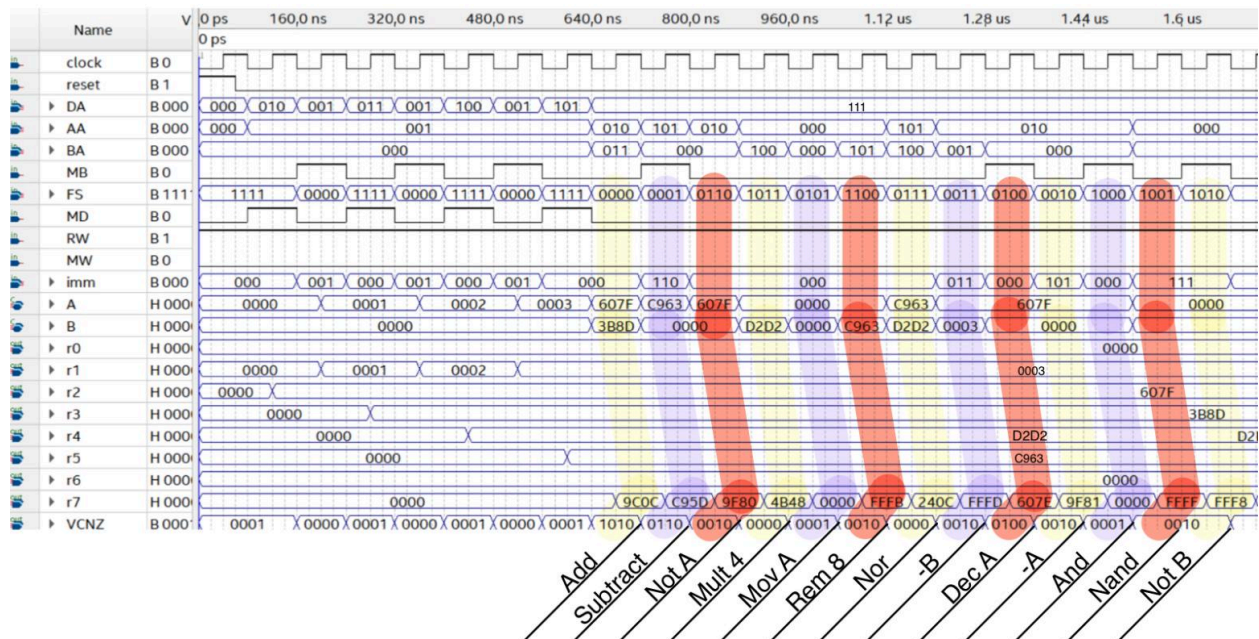


Figure 1.8 Annotated Simulation

The annotated simulation above, has the trail from FS code through imm, A, B, result, and status bits highlighted for a single operation so they can each be seen easily due to the slight shift.

Meanwhile, the simulation below is the regular simulation output

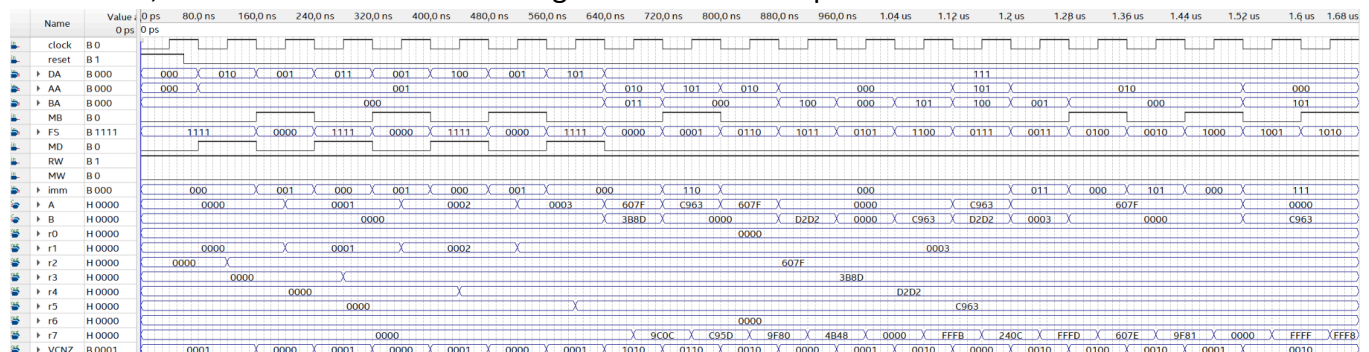


Figure 1.9 Simulated Waveform